A Dual Attention Network for Multimodal Remote Sensing Image Matching

Kaiyang Han
National University of Defense Technology (NUDT)
Changsha, China
hkynudt@163.com

Fanzhi Cao
National University of Defense Technology (NUDT)
Changsha, China
513572289@qq.com

Tianxin Shi
National University of Defense Technology (NUDT)
Changsha, China
tianxin_shi@126.com

Pu Wang*
National University of Defense Technology (NUDT)
Changsha, China
wp421189@nudt.edu.cn

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***Abstract*—Multimodal remote sensing image matching is a crucial task for many remote sensing applications. Due to different imaging mechanisms and conditions, multimodal images often suffer from significant nonlinear radiation distortions (NRD). To address this issue, this paper proposes a robust feature matching framework, where the learnable descriptors can be invariant to contrast changes. Firstly, we created a multimodal remote sensing image dataset for training the algorithm, as there is a lack of datasets in this field. Secondly, we use the phase congruency algorithm to preserve the invariant features between multimodal image pairs and optimize the parameters to make them more suitable for multimodal remote sensing image matching tasks. Finally, we design a matching network that introduces both position attention and channel attention modules to model the spatial and dimensional dependencies. Experimental results show that our method exhibits significant advantages over popular algorithms on publicly available datasets.**

Keywords：multimodal image matching; remote sensing; phase congruency; attention mechanism

I. INTRODUCTION

Image matching is a challenging visual task that involves identifying identical or similar structures or content from two images and establishing correspondences[1]. It is a crucial step for subsequent tasks such as image fusion, change detection, and SLAM. Multimodal remote sensing images refer to remote sensing images captured under different sensors or different shooting conditions. These images often suffer from significant nonlinear radiation distortions, which pose great difficulties for accurate matching[2].

Traditional matching methods are mainly applied to visible light images, which produce a large amount of noise when facing multimodal images, and often fail to find enough correct matching points to complete the matching task. Currently, multimodal remote sensing image matching faces the following main challenges: Firstly, the field of multimodal remote sensing image matching has not yet seen large-scale datasets like those in the field of multimodal medical image matching due to the difficulty in data collection. Deep learning is data-driven, and the quality and quantity of data directly affect the final results. The lack of datasets limits the application of deep learning in multimodal remote sensing image matching. Secondly, due to differences in imaging principles and imaging conditions, there are significant NRD between remote sensing images. The current common practice is to preserve the common structural features using phase consistency[3] and then perform feature extraction. However, phase consistency is not designed for multimodal remote sensing images and has numerous adjustable parameters. Existing algorithms generally use default parameters, so parameter optimization is needed to make it more suitable for multimodal remote sensing image matching tasks. Thirdly, the feature descriptors of traditional methods are hand-crafted and are generally designed based on grayscale or gradient information, which cannot capture deeper and more abstract features. When faced with multimodal images with significant NRD, this can lead to a large amount of noise that misleads matching and results in many outliers. Convolutional neural networks (CNN) are known for their robust feature extraction capabilities and have proven to be effective in various computer vision domains such as image classification, object detection, and semantic segmentation. Despite their success in these fields, CNN have not yet established a dominant position in multimodal remote sensing image matching task[4].

To address the aforementioned challenges, we created a dataset and optimized the parameters of the phase congruency algorithm, and designed a network matching. The pipeline of our proposed algorithm is illustrated in the figure 1, which consists of three stages: feature detection, descriptor extraction, and feature matching.

In the feature detection stage, we used the optimized phase congruency algorithm to preserve the invariant features of both the reference and sensed images, and then utilized the FAST algorithm to extract the invariant feature points. These feature points are then used to extract feature descriptors in the descriptor extraction stage. Finally, in the feature matching stage, we employ the nearest neighbor strategy to match the descriptors.

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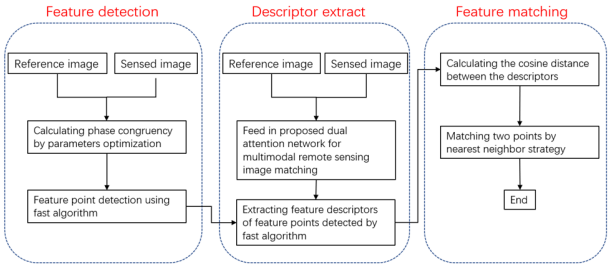


Figure 1. The pipeline of our method

The matching network we proposed is shown in the figure 2. The network takes the image to be matched and processes it using a multi-layer neural network to obtain a feature map that is 1/8th of the original size and has 128 dimensions. The feature map is then passed through two parallel attention modules: the position attention module (PAM) and the channel attention module (CAM). The PAM models the spatial dependency between any two positions in the feature map using self-attention mechanism, while the CAM models the dependency relationship between any two channels and updates each channel mapping by weighted summing all channel mappings. The outputs of the two modules are fused to further enhance the feature representation. Finally, the feature map is upsampled to the original size by bilinear interpolation, and the vector along the channel dimension of the feature point is extracted as the feature descriptor. It is important to note that the same network structure and parameters are used for both the reference image and the sensed image.

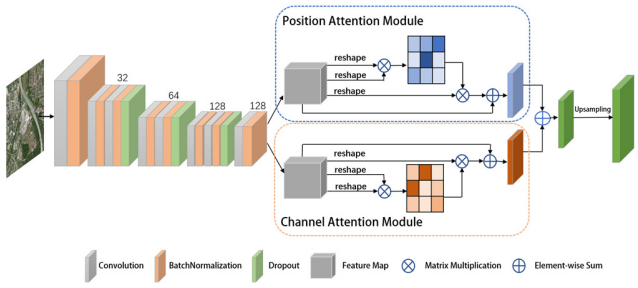


Figure 2. The network of proposed

Based on the above analysis, this paper's main contributions are summarized as follows:

1. Given the scarcity of datasets in the field of multimodal remote sensing image matching, we collected a large number of images from public sources. To ensure the algorithm's robustness, we selected images that included cloud scenes and were taken in different seasons and years.
2. We studied and optimized the parameters of phase congruency for multimodal remote sensing image matching. We conducted extensive experiments to investigate the effects of different parameters on the algorithm and achieved better results with the optimized parameters than with the default ones.
3. We proposed a network with dual attention mechanisms for multimodal remote sensing image matching. The two attention modules model spatial and channel dependencies, respectively, and improve the matching performance.

II. Ralated Work

There are three main types of methods for multimodal remote sensing image matching: region-based, feature-based, and learning-based[5]. Region-based matching methods face challenges in establishing a reliable similarity measurement approach, with traditional spatial domain methods such as Sum of Squared Differences (SSD), Normalized Cross-Correlation (NCC), and Mutual Information (MI) having high time complexity. Later, some frequency domain methods appeared, mainly based on the idea of converting two images with different modalities to the frequency domain for matching. Xiang et al. proposed the OS-PC algorithm[6], which combines robust feature representation with three-dimensional phase correlation. HOPC descriptor, which uses phase congruency features, has also shown better matching performance for images with significant NRD[7]. Region-based matching methods can only handle simple translation transformations, so their application range is limited.

The feature-based matching method aims to correspond and match two images by identifying their invariant features. Scale-invariant feature transform (SIFT) is widely used for optical image matching[8]. However, the direct application of the SIFT algorithm to multimodal remote sensing image matching greatly reduces its effectiveness. Based on the SIFT algorithm, Ma et al. proposed the PSO-SIFT algorithm[9], which integrates the spatial position, scale, and direction information of the feature points, achieving good results in matching visible light and SAR images. Ye et al. proposed using MMPC-Lab for feature detection and local histogram of orientated phase congruency (LHOPC) to describe features, which can resist changes in illumination and contrast[10]. The radiation-invariant feature transform (RIFT) first proposed the method of constructing a maximum index map to resist NRD[11]. Yao et al. extends the phase consistency model by introducing an anisotropic weighted moment equation. Fan et al. proposed a coarse-to-fine matching strategy (3MRS)[12], which further improves matching accuracy with a three-dimensional phase correlation strategy based on the maximum index map. The feature-based matching method has certain robustness, making it widely used.

At present, deep learning has also made some progress in the field of image matching. MatchNet proposes to use two networks (Siamese networks) with the same parameters and structure to extract features for image matching[13]. DFM extracts features for matching using existing pre-trained networks[14]. Learning-based methods demonstrate significant advantages due to their powerful feature extraction ability. But as mentioned above, due to the lack of large-scale data sets, learning-based methods have not achieved a leading position

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III. METHOD

A. Phase Congruency Parameter Optimization

Phase congruency refers to the points in a signal where the phase components are consistent. This method provides a measure that is independent of the overall signal amplitude and is therefore less affected by changes in image contrast or brightness. In this paper, we utilize the excellent radiation robustness of phase congruency for the task of multimodal image matching. Feature point extraction is performed by constructing a phase consistency map and replacing the original features. This preserves the invariant structural features of the multi-modal remote sensing image pair, which is beneficial for extracting invariant feature points. We use Log-Gabor filters to calculate phase consistency at multiple scales and orientations, as shown in the following equation:

$$PC(x,y) = \frac{\sum_s \sum_o w_o(x,y)\lfloor A_{so}(x,y)\Delta\Phi_{so}(x,y) - T \rfloor}{\sum_o \sum_s A_{so}(x,y) + \varepsilon} \quad (1)$$

PC(x, y) represents the magnitude of phase consistency at point (x, y), where wo is the weight factor of frequency distribution, $A_{so}(x, y)$ is the amplitude at position (x, y) on the wavelet scale s and direction o. ε is a small constant to prevent the denominator from being zero. The $\lfloor \cdot \rfloor$ operator means that when the result of the operator inside is positive, the result of the operator is equal to the result inside the operator; otherwise, the result is 0. Therefore, this operation ensures that the PC result is non-negative. T is the noise intensity, and only energy values greater than the noise power are included in the result. $\Delta\Phi_{so}$ is a sensitive phase deviation function defined as:

$$A_{so}(x,y)\Delta\Phi_{so}(x,y) = \left(e_{so}(x,y)\overline{\phi_e}(x,y) + r_{so}(x,y)\overline{\phi_d}(x,y)\right) - \left(e_{so}(x,y)\overline{\phi_d}(x,y) - r_{so}(x,y)\overline{\phi_e}(x,y)\right) \quad (2)$$

$$\overline{\phi_e}(x,y) = \sum_s \sum_o e_{so}(x,y) / E(x,y) \quad (3)$$

$$\overline{\phi_d}(x,y) = \sum_s \sum_o s_{so}(x,y) / E(x,y) \quad (4)$$

Where $e_{so}(x, y)$ and $s_{so}(x, y)$ represent the convolution of the image with even-symmetric and odd-symmetric Log-Gabor wavelets, respectively, to obtain the response at scale s and orientation o. E(x, y) is the local energy function, obtained by convolution with a pair of orthogonal filters:

$$E(x,y) = \sqrt{(\sum_s \sum_o e_{so}(x,y))^2 + (\sum_s \sum_o r_{so}(x,y))^2} \quad (5)$$

However, the phase congruency algorithm was not originally proposed for the task of multimodal image matching, and an important factor, the optimization and fine-tuning of control parameters, was often overlooked in previous literature. To improve the robustness of phase consistency to nonlinear radiation differences, this paper studies a set of optimal parameters as control parameters for the algorithm pipeline.

Since deep learning requires a lot of time for training, validating the goodness of parameters throughout the entire algorithm can take a lot of time. Therefore, we propose a Coarse-to-Fine parameter optimization strategy. We first use the Structural Similarity (SSIM) and FAST algorithms to conduct a coarse screening of the parameters and then conduct fine screening in our deep learning algorithm to select the optimal parameters. We measure the structural similarity between the phase congruency maps PC_a of the reference image and PC_b of the sensed image using the SSIM and measure the number of detected feature points using the FAST algorithm. We believe that the larger the SSIM and the number of feature points between PC_a and PC_b, the better the parameter set.

The phase consistency algorithm has eight control parameters, namely [N_s, N_o, λ_{min}, η, σ, k, C_o, g]. Their default values, ranges, and parameter meanings are shown in the table I. Among them, Ns = 4 and No = 6 have been proven to be the optimal parameters , and the variations of λ_{min}, k, Co, and g have little impact on the results of the phase consistency algorithm, as demonstrated in our study. Therefore, we use default values for these parameters, and the two parameters that mainly affect the algorithm results are η and σ[15].

TABLE I. PHASE CONGRUENCY PARAMETERS

Parameters	Default	Range	Meaning
N_s	4	[3,6]	Number of wavelet scales
N_o	6	[1,6]	Number of filter orientations
λ_{min}	3	⩾3	Wavelength of smallest scale filter
η	2.1	[1.3,1.6,2.1,3]	Scaling factor between successive filters
σ	0.55	[0.1~1]	Ratio of the standard deviation
k	2	[10~20]	Noise scaling factor
C_o	0.5	(0~1]	The fractional measure of frequency spread
g	10	[1~50]	The control coefficient of the sigmoid

By adjusting the two parameters, as shown in figure 3, we found that when σ is less than 0.3, the PC map will produce a lot of noise, and the invariant structural features will be submerged in the noise. The SSIM of the image pair will be greatly reduced, and a large number of noise points will be detected as feature points. When σ is greater than 0.7 or η is greater than 2.1, the PC map will become very dim, unable to preserve the invariant features of the image, and the number of detected feature points will be too small.

Based on the above analysis, there are two options for η, each with seven combinations by changing the value of σ within the range of (0.3, 0.7) at 0.05 intervals. Therefore, we determine the optimal parameters by comparing the performance of 14 parameter combinations in the network proposed in this paper.

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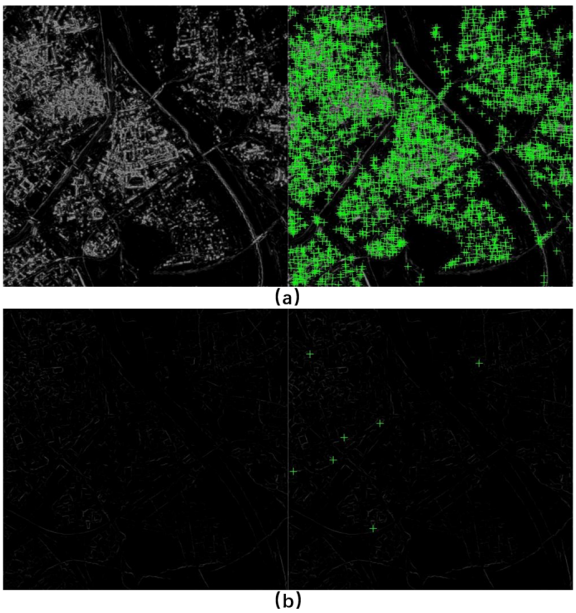


Figure 3. (a)show σ is less than 0.3.(b)show σ is greater than 0.7 or η is greater than 2.1

B. Position Attention Module

To construct a more matchable feature descriptor, we introduced the PAM to aggregate contextual semantic information. The module encodes more extensive contextual information as local features, thereby enhancing the feature expression ability of the descriptor and making it more specific and matchable. The construction process of the module is described below.

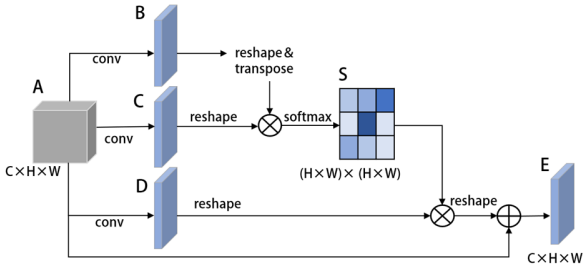


Figure 4. PAM

As shown in the above figure 4, we first use a convolutional layer to generate three new feature maps B, C, and D from feature map A. Like A,$\{B, C, D\} \in R^{C\times H\times W}$. Then we reshape them into $R^{C\times N}$, where $N = H \times W$ is the number of pixels. Next, we multiply the transpose of B with C to obtain a matrix, which is then passed through a softmax layer to obtain the position attention map $S \in R^{N\times N}$:

$$S_{ij} = \frac{exp(B_i \cdot C_i)}{\sum_{i=1}^{N} exp(B_i \cdot C_i)} \tag{6}$$

Where S_{ij} represents the influence of the jth position on the ith position, and the more correlated they are, the higher score S_{ij} will achieve. Next, we reshape D into $R^{C\times N}$ and perform matrix multiplication with S. The resulting matrix is then reshaped into $R^{C\times H\times W}$ and multiplied by a scaling factor α. Finally, the output is obtained by adding this product to the feature map A, resulting in the final output $E \in R^{C\times H\times W}$:

$$E = \alpha \cdot reshape\{DS\} + A \tag{7}$$

The parameter α is a learnable parameter. From the above two formulas, we can see that the weight of each position in E is the weighted sum of all position features and the original features, thus possessing a global contextual view. Additionally, the location attention map is used to selectively aggregate features from different positions, leading to descriptors with greater specificity. The feature representation between different keypoints will be more distinguishable with the support of global semantic information, resulting in improved matching performance.

C. Channel Attention Module

Our descriptor consists of the channel direction of the feature map, with a total of 128 dimensions. Different channel feature maps are mutually correlated and influential, representing features at different levels. To better optimize the feature descriptor, we introduced a CAM to explicitly model the interdependence between channels.

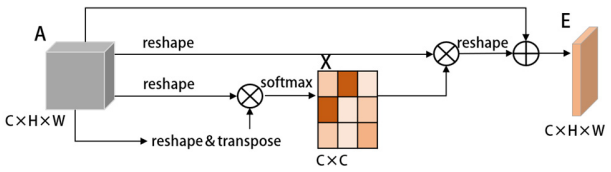


Figure 5. CAM

The channel attention map is shown in the above figure 5. First, we reshape the feature map $A \in R^{C\times H\times W}$ to $A \in R^{C\times N}$, then we perform matrix multiplication between A and its transpose. After passing through a softmax layer, we obtain the channel attention map $X \in R^{C\times C}$:

$$X_{ij} = \frac{exp\{A_i \cdot A_j\}}{\sum_{i=1}^{C} exp\{A_i \cdot A_j\}} \tag{8}$$

Where X_{ij} represents the impact of the jth channel on the ith channel. Next, we reshape A into $A \in R^{C\times N}$ and perform matrix multiplication with the channel attention map $X \in R^{C\times C}$. The resulting matrix is then reshaped into $R^{C\times H\times W}$. Finally, we multiply the result by a scale parameter β and add it pixel-wise to A to obtain the final result $E \in R^{C\times H\times W}$:

$$E = \beta \cdot reshape\{XA\} + A \tag{9}$$

Where β is a learnable parameter. From the above two formulas, we can see that the final feature of each channel is a weighted sum of the original channel and all other channels,

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which models the feature dependency between channels and helps to enhance the discriminability of feature descriptions.

IV. Experiments

This paper uses a self-made dataset for training and the multi-modal image matching dataset proposed by Ma et al. for testing. The dataset includes multi-modal images from three different domains: remote sensing, computer vision, and medical imaging, with the remote sensing domain being the main focus of this study. The network training was performed on a single 24GB NVIDIA 3090 GPU.

To validate the effectiveness of the our algorithm, four algorithms were selected for comparison: RIFT, HAPCG, 3MARS, and DFM, where DFM is a deep learning method that does not require training according to its paper, and thus was not retrained on our dataset.

A. Evaluation Metrics

This paper primarily employs three evaluation metrics to assess the performance of the proposed algorithm: the number of correctly matches (NCM), matching accuracy (ACC), and matching runtime (RT). NCM refers to the number of predicted matching points that are within five pixels of the true matching points. ACC refers to the percentage of correctly matched points out of the total matching points. RT refers to the time required for the algorithm to complete the matching process.

B. Parameter Optimization

Based on the method described above, we obtained a preliminary optimal range of parameters. Next, we input these parameters into the neural network to obtain the matching results on the test set. The main comparison of these parameters was based on ACC. The results are shown in the table II and figure 6.

TABLE II. PARAMETER OPTIMIZATION RESULTS

setting of η	**The percentage of ACC(%) (setting of σ)**						
	0.35	***0.4***	***0.45***	***0.5***	***0.55***	***0.6***	***0.65***
1.3	51.26	53.76	53.15	52.07	51.83	51.59	50.68
1.6	53.43	**55.25**	53.67	52.72	52.34	52.22	51.28

As shown in the above table II, the best matching accuracy is achieved when η=1.6 and σ=0.4. Unless otherwise specified, this parameter will be used in the following experiments.

C. Contrast Test

The multi-modal remote sensing dataset proposed by Ma et al. can be divided into seven specific types, including CrossSeason, Day-Night, Depth-Optical, Infrared-Optical, Map-Optical, Optical-Optical, and Sar-Optical. We will compare the performance of the algorithm on each of these types. Figure 7 shows some matching results of multimodal remote sensing image pairs.

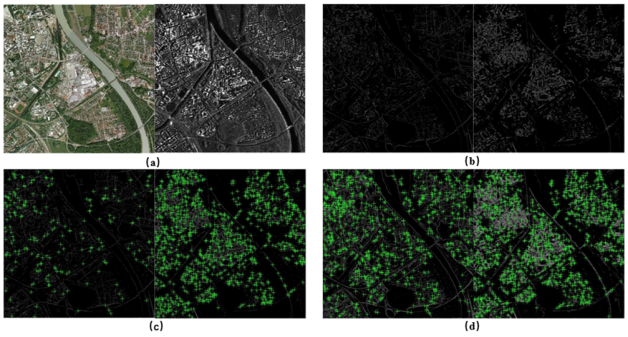


Figure 6. The results of PC maps .（a）Optical and SAR image pair. (b)PC maps obtained by default parameters. (c)feature points detected in (b).(d)show more high-quality feature points can be detected after parameter optimization

TABLE III. RESULTS OF ACC

Method	**ACC（%）**						
	CrossSeason	***Day-Night***	***Depth-Optical***	***Infrared-Optical***	***Map-Optical***	***Optical-Optical***	***Sar-Optical***
RIFT	12.6	20.16	30.48	35.05	28.85	34.04	34.41
HAPCG	6.93	5.24	6.18	13.43	9.47	7.95	5.15
3MARS	13.09	20.11	31.68	36.11	31.47	33.58	33.70
DFM	24.66	40.00	**61.77**	20.83	**44.29**	**80.27**	71.16
OURS	**27.46**	**43.52**	55.63	**47.21**	35.74	60.13	**73.65**

TABLE IV. RESULTS OF NCM

Method	**NCM**						
	CrossSeason	***Day-Night***	***Depth-Optical***	***Infrared-Optical***	***Map-Optical***	***Optical-Optical***	***Sar-Optical***
RIFT	128	267	393	352	392	469	390
HAPCG	201	174	271	**471**	403	256	211
3MARS	125	259	488	347	**488**	446	352
DFM	18	1	10	3	2	146	12
OURS	**345**	**499**	**853**	435	356	**931**	**485**

TABLE V. RESULTS OF RT

Method	**RT(S)**						
	CrossSeason	***Day-Night***	***Depth-Optical***	***Infrared-Optical***	***Map-Optical***	***Optical-Optical***	***Sar-Optical***
RIFT	9.52	10.48	11.10	7.32	11.65	9.98	8.57
HAPCG	7.34	5.00	6.12	4.63	6.26	4.46	5.60
3MARS	9.94	10.17	12.79	6.79	13.05	9.92	7.96
DFM	**1.38**	1.30	1.58	1.52	1.84	1.73	1.96
OURS	2.31	**1.14**	**1.35**	**1.16**	**1.28**	**1.17**	**1.43**

From Table III, it can be seen that our method demonstrates the best matching accuracy on multiple types of data, but performs worse than DFM on several data types. Table IV shows the comparison results of different algorithms in terms of the number of correctly matched points. It can be observed that our algorithm outperforms other algorithms in the vast majority of data types, while the DFM algorithm obtains very few correctly matched points on quite a few data types, which could lead to unsuccessful matching tasks. Table V shows the running time of different algorithms. Our method is generally more than four times faster than traditional methods such as RIFT, HAPCG, and 3MARS. Compared with DFM algorithm which also employs deep learning, our method is also leading in the vast majority of data types.

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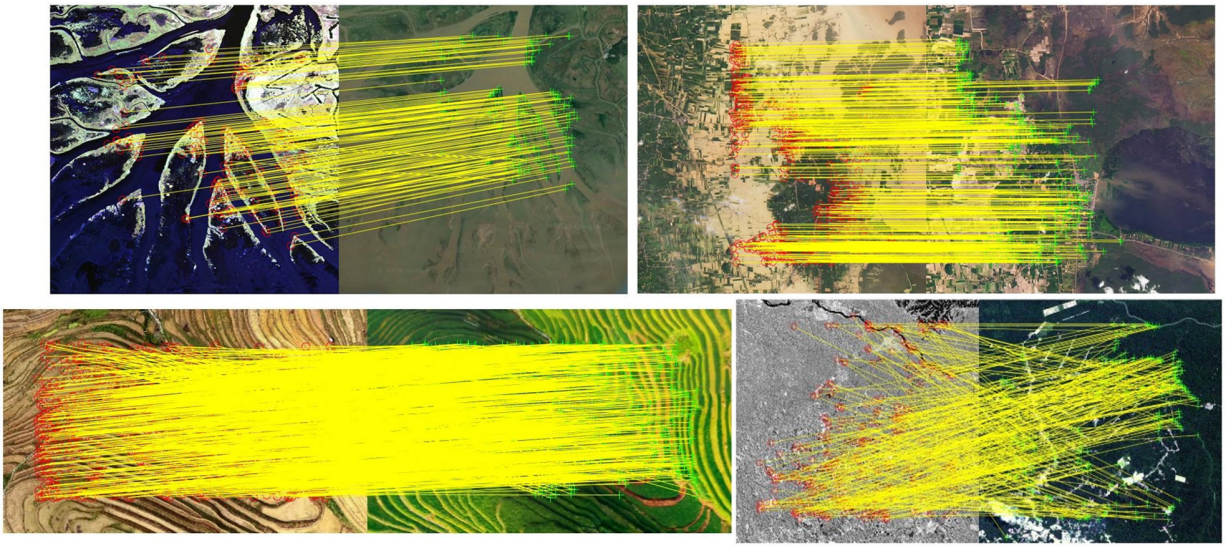


Figure 7. Some matching results of our algorithm

D. Ablation Experiment

We further investigated the optimization of phase consistency parameters and the impact of two attention modules on the matching results, and conducted ablation experiments. In these experiments, "OP" represents the optimal parameters obtained using phase consistency, "PAM" represents the position attention module, and "CAM" represents the channel attention module.

TABLE VI. RESULTS OF ABLATION EXPERIMENT

OP	PAM	CAM	ACC
			51.23%
√			53.24%
√	√		53.58%
√		√	54.00%
√	√	√	55.25%

From the table VI, it can be seen that the accuracy is significantly improved after parameter optimization. When using the optimal parameters, using PAM or CAM alone can improve performance. The best performance is achieved when both modules are used simultaneously. The results of the ablation experiment show that our strategy can improve the performance of matching on the basis of the basic network.

V. CONCLUSION

In this paper, we propose a robust algorithm for multimodal remote sensing image matching. To support our research, we collect and create a dataset using open sources. Our algorithm employs a coarse-to-fine strategy to obtain optimal phase consistency parameters for multimodal remote sensing image matching tasks. We introduce a dual attention network for image matching, which includes a position attention module (PAM) and a channel attention module (CAM). These modules model spatial and channel dependencies, respectively, to enhance the specificity of the feature descriptor. Our experiments demonstrate that the proposed algorithm performs well on public datasets and is competitive with several popular algorithms. In future work, we aim to optimize the algorithm's performance in scenes involving rotation and scale of images.

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